LESSFormer: Local-Enhanced Spectral-Spatial Transformer for Hyperspectral Image Classification

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***Abstract*— Currently, the convolutional neural networks (CNNs) have become the mainstream methods for hyperspectral image (HSI) classification, due to their powerful ability to extract local features. However, CNNs fail to effectively and efficiently capture the long-range contextual information and diagnostic spectral information of HSI. In contrast, the leading-edge vision transformers are capable of capturing long-range dependencies and processing sequential data such as spectral signatures. Nevertheless, preexisting transformer-based classification methods generally generate inaccurate token embeddings from a single spectral or spatial dimension of raw HSIs and encounter difficulty modeling locality with insufficient training data. To mitigate these limitations, we propose a novel local-enhanced spectral–spatial transformer (i.e., LESSFormer) method specifically devised for HSI classification. Two effective and efficient modules are designed in LESSFormer, i.e., the HSI2Token module and the local-enhanced transformer encoder. The former is devised to transform HSI into the adaptive spectral–spatial tokens and the latter is built to further enhance the representation ability of tokens by reinforcing the local information explicitly with a simple attention mask as well as retaining the long-range information in the meantime. Extensive experimental results on the new Xiong'an dataset and the widely used Pavia University and Houston University datasets have shown the superiority of LESSFormer over other state-of-the-art networks.**

***Index Terms*— Hyperspectral image (HSI) classification, local contextual information, spectral partition (SP), superpixel segmentation, transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) usually contain hundreds of images at different wavelengths depicting the same scene, where the abundant spectral and spatial information are conducive to distinguishing materials through subtle discrepancies in spectral reflectance and spatial patterns [1]. Thus, HSIs have been exploited in numerous applications over the years, such as precision agriculture, military reconnaissance, and mineral mapping [2]–[5]. As a crucial processing procedure, HSI classification lays the foundation for these applications, aiming to identify the land cover types of each pixel in HSIs.

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In the early stages, how to exploit the high-dimensional spectral vectors for HSI classification has been widely explored. A large number of spectral classifiers including support vector machine (SVM), k-nearest neighbor (KNN), and random forest (RF) were directly applied to predict every pixel of HSIs. Nevertheless, conventional spectral classifiers encounter the difficulty of "curse of dimensionality" resulting from high-dimensional data [6]. Therefore, some researchers have focused on combining machine learning classifiers with dimension reduction methods, which can be divided into linear techniques such as principal component analysis (PCA) and linear discriminant analysis (LDA) [7], and nonlinear techniques such as locality preserving projections (LPPs) [8] and kernel Fisher discriminant analysis (KFDA) [9]. Duan *et al.* [10] developed an advanced nonlinear method adopting the geodesic-based sparse manifold hypergraph to extract the low-dimensional representations, which were fed into classifiers such as SVM for classification. In [11], a novel nonlinear dimension reduction method was designed considering multiple properties of HSI to capture the discriminative features, and the extracted features were classified using conventional classifiers. Spectral partition (SP) is another alternative that processes high-dimensional hyperspectral data without losing spectral information, based on the fact that spectral signature is obtained by combining signals from different instruments and presents the properties of grouped features [12]. Li *et al.* [13] applied the SP operation to divide raw HSI into several groups of nonoverlapping subbands and then processed these groups individually, which achieved decent performance using small training samples. Due to the spectral variability in raw HSI, these methods that only consider the spectral information still yield unsatisfactory classification results. Recent studies have indicated that spatial contextual information is beneficial for boosting classification performance [14]–[16]. As a result, an increasing number of spectral–spatial classification approaches were proposed by incorporating handcrafted spatial features including invariant attribute profiles [14] and Gabor filterings [17] into the spectral classifiers, which significantly improved the classification accuracy. In addition, superpixel segmentation has been extensively exploited as a preprocessing step to preserve the spatial structure [18]–[20]. However, these spectral–spatial classification methods heavily rely on expert knowledge and fail to capture the latent high-level representation of HSIs.

More recently, deep learning (DL) has achieved remarkable breakthroughs in various computer vision (CV) and natural language processing (NLP) tasks, thus stimulating interest in

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DL across a wide range of fields. The initial DL attempts in HSI classification [21], [22] adopted the stacked autoencoder (SAE) and deep belief network (DBN), which, however, cannot retain the spatial features efficiently with 1-D vectors as input. Subsequently, several prominent DL models have served as backbones for HSI classification, including the sequence models such as recurrent neural networks (RNNs) [23], [24], generative models such as generative adversarial networks (GANs) [25], [26], as well as other advanced models such as fully convolutional networks (FCNs) [27], [28] and graph neural networks (GNNs) [29], [30]. Due to the natural tendency to encode spatial contextual information, convolutional neural networks (CNNs) have become the mainstream for HSI classification. Numerous CNN-based classification methods have been developed from 1-D CNNs to 3-D CNNs. In [31], Li *et al.* used the pixel-pair features extracted by 1-D CNN to explore the correlation between pixels, where the convolution operation was performed in the spectral domain. Zhao and Du [7] fed a few principal component bands of HSI into a five-layer 2-D CNN for spectral–spatial classification. To exploit the inherent 3-D structure of HSI, a 3-D CNN framework was proposed to jointly process the spectral and spatial features in [32]. Apart from the single CNNs, a hybrid CNN was designed in [33], which can extract and refine the spectral–spatial information by combining the 2-D CNN and 3-D CNN. Moreover, several studies incorporated residual connections, dense connections, and feedback connections to construct deeper and wider CNNs for HSI classification [34]–[36]. In contrast, a few works turned their attention to lightweight CNNs based on the modified convolutional layer such as depth-wise separable convolution. In order to acquire more contextual information, the attention mechanism has also been integrated with CNNs for HSI classification [37]–[39]. Although multifarious CNN-based classification methods have achieved promising accuracy, they still suffer from two shortcomings. For one thing, it is not trivial for CNNs to effectively and efficiently capture the long-range contextual information from the cropped HSI patches with the small convolutional kernel. For another, CNNs slide over the spatial dimension of HSI and treat the channel-wise features equally, thus losing the diagnostic features in the spectral domain to some extent.

The advanced vision transformer (ViT) [40] and its improved variants [41]–[43] have emerged as alternative backbones in many vision tasks, which are purely composed of the self-attention (SA) mechanism. Significantly different from the CNN architecture, the effective receptive fields of ViT become considerably global in the shallow layer, while the receptive fields of CNN counterparts grow gradually, which illustrates the superior ability of ViT to model the long-range dependencies [44]. In addition, owing to the strong capability to process sequential data, ViT is able to extract discriminative spectral features even confronted with high spectral variability in complex HSI scenarios. Therefore, a few works related to ViT have been devised for HSI classification [45], [46]. Hong *et al.* [47] applied ViT to HSI classification by regarding the spectral signature as sequential data and feeding the preprocessed local spectral embeddings (a.k.a., spectral tokens) to the transformer encoder with cross-layer skip connections, which can retain the local spectral representation and maximize the information reuse among network layers. While in [48], a transformer-based classification method with good generalization capability was developed by exploiting diverse region-based spatial embeddings (a.k.a., spatial tokens) and the bidirectional encoder representations from transformers (BERT) model. Instead of generating token representations directly from the original HSI, Sun *et al.* [49] designed an efficient CNN to extract shallow spectral–spatial features prior to the tokenization step, which helps to generate more accurate token representations. Nevertheless, there exist two limitations of these transformer-based classification methods. On the one hand, the preexisting transformer-based methods perform tokenization without taking full consideration of the characteristic of HSIs. Compared with natural images (i.e., red, green, blue (RGB) images), HSIs are the image–spectrum merging cube data with rich and even redundant information. Therefore, it is not appropriate to generate tokens from a single dimension (i.e., the spectral dimension or spatial dimension) of the raw HSIs. On the other hand, despite the fact that the global SA module enables ViT to effectively model long-range dependencies, they ignore the spatial prior knowledge that near pixels tend to be more related than distant pixels. Thus, a large amount of training data is generally required for ViT to implicitly learn the inductive bias in modeling locality, which is undesirable for HSI classification tasks with limited labeled data. Moreover, irrelevant relations will inevitably be considered in the calculation of global attention, which introduces noise to semantic modeling and results in performance degradation.

In view of these problems, we propose a novel local-enhanced spectral–spatial transformer (i.e., LESSFormer) method dedicated to HSI classification, which transforms HSI into adaptive spectral–spatial token representations and further promotes the discriminative capability of these tokens by simultaneously capturing the local and long-range information. First, in order to extract the discriminative low-level spectral–spatial features and minimize image redundancy, an HSI2Token module is constructed to convert an HSI into adaptive spectral–spatial tokens. Second, a novel local-enhanced transformer is utilized to further enhance the discriminative ability of token embeddings, which explicitly emphasizes the local features by imposing the local spatial prior that restricts attention to a local neighborhood region and meanwhile preserves the long-range features. At last, to predict the label of each pixel in HSI, a simple classification head is employed to project the enhanced tokens back into the pixel-level spectral–spatial features. In general, by integrating the image redundancy property, image-spectrum merged characteristic, and local spatial prior knowledge of HSI into the network architecture design, the LESSFormer method aims at fostering strengths and circumventing weaknesses of vision transformer to face the challenges in HSI classification task, in order to obtain competitive classification performance.

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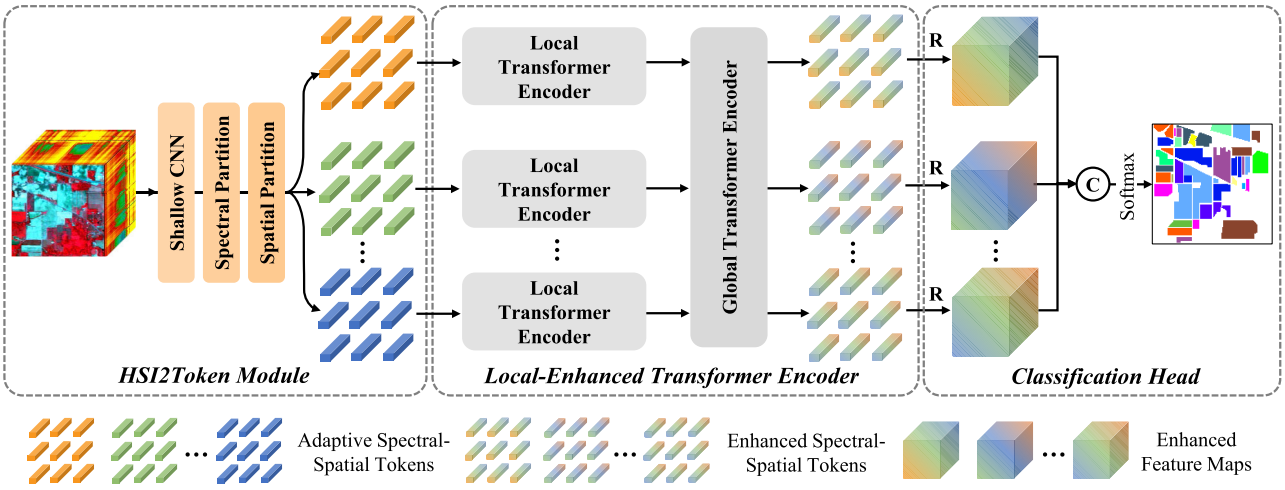


Fig. 1. Overall framework of LESSFormer. The LESSFormer framework consists of three modules: HSI2Token module, local-enhanced transformer encoder, and classification head.

The main contributions of this article are as follows.

1) We develop an LESSFormer for HSI classification, with full consideration of the image redundancy property, image-spectrum merged structure, and spatial prior knowledge of HSIs.
2) We design two effective and efficient modules in LESSFormer, i.e., the HSI2Token module and local-enhanced transformer encoder, to generate representative low-level spectral–spatial tokens tailored for HSI data and enhance the discriminative ability of these tokens by capturing the local and long-range dependencies simultaneously with the aid of a simple attention mask.
3) Extensive experimental results on a new HSI dataset [i.e., Xiong'an (XA)] and two widely used HSI datasets [i.e., Pavia University (PU) and Houston University (HU)] illustrate that the proposed LESSFormer outperforms other state-of-the-art networks.

The remaining of the article is organized as follows. Section II revisits the related work on ViT. Section III illustrates the proposed LESSFormer in detail. Section IV presents extensive experimental results and comprehensive analyses. Section V draws the conclusions of the article and provides plausible future research work.

II. Revisiting Vision Transformer

Motivated by the remarkable success of transformers in NLP, emerging works based on transformers are introduced for various CV tasks. ViT pioneered applying Transformer [50] to a sequence of image patches and achieved a breakthrough. Concretely, ViT includes three steps [51]: First, perform tokenization that splits images into nonoverlapping patches to resemble tokens in NLP. Second, apply the pure transformer encoder to tokens. Third, conduct classification using the "classification token" that corresponds to the global image representation. The transformer encoder is the essential part of ViT, which involves alternating layers of multihead self-attention (MSA) and multilayer perceptron (MLP). The layer normalization (LN) [52] and residual connection are employed before and after each layer, and the Gaussian Error Linear Unit (GELU) activation function [53] is applied in MLP, which can be formulated as

$$\begin{aligned} \mathbf{T}' &= \mathbf{T} + \mathrm{MSA}(\mathrm{LN}(\mathbf{T})) \\ \mathbf{T}_{\mathrm{attn}} &= \mathbf{T}' + \mathrm{MLP}\big(\mathrm{LN}\big(\mathbf{T}'\big)\big) \end{aligned} \tag{1}$$

where $\mathbf{T} \in \mathbb{R}^{n\times d}$ is a sequence of tokens, $\mathbf{T}_{\mathrm{attn}} \in \mathbb{R}^{n\times d}$ is a sequence of enhanced tokens. The n and d represent the length of the sequence and the feature dimension, respectively. The MSA denotes the multihead self-attention layer, LN denotes the layer normalization layer, and MLP represents the multilayer perceptron layer. The layer l is omitted in (1) for the sake of simplicity.

Among these building blocks, SA layer plays a crucial role in building long-range dependencies to enhance features, acting as a weighted sum over the original features. In the SA layer, the input tokens $\mathbf{T}$ are first linearly projected into the queries $\mathbf{Q} \in \mathbb{R}^{n\times d}$, keys $\mathbf{K} \in \mathbb{R}^{n\times d}$, and values $\mathbf{V} \in \mathbb{R}^{n\times d}$, respectively, then the enhanced representations are generated by a weighted average over values as follows:

$$\begin{aligned} \mathbf{A} &= \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right) \\ \hat{\mathbf{V}} &= \mathbf{A}\mathbf{V} \end{aligned} \tag{2}$$

where $\mathbf{A} \in \mathbb{R}^{n\times n}$ is the attention score matrix, and $\hat{\mathbf{V}} \in \mathbb{R}^{n\times d}$ denotes the matrix of enhanced values.

MSA extends SA by dividing queries, keys, and values into h (i.e., number of heads) parts to conduct h SA operations in parallel, then performing a linear transformation on the concatenated outputs, which can jointly capture dependencies from several different subspaces. An extra "classification token" is attached and the position embedding is added to the input tokens prior to performing the first MSA operation. The interested readers can refer to [40] for more details.

It is worth noting that a few vision transformer-based methods have been applied for HSI classification [47]–[49], which generally generate tokens in a single dimension of the original high-dimensional HSI and encounter difficulties capturing local information among tokens. Considerably different from these methods, the proposed LESSFormer method

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TABLE I
DEFINITIONS OF THE IMPORTANT NOTATIONS

Module	Notation	Definition
HSI2Token Module	$\mathbf{H}$	Original HSI data
	$\mathbf{X}$	Low-level spectral-spatial features
	$\mathbf{R}$, $\hat{\mathbf{R}}$	Original, column-normalized pixel-superpixel association matrices
	$\mathbf{X}_{\text{spe}} = [\mathbf{X}^1, \mathbf{X}^2, \dots \mathbf{X}^G]$	Output features of spectral partition
	$\mathbf{X}^i_{\text{spa}} = [\mathbf{X}^{i,1}, \mathbf{X}^{i,2}, \dots \mathbf{X}^{i,M}]$	Output features of spatial partition in one group
	$\mathbf{Y}^i = [\mathbf{y}^{i,1}, \mathbf{y}^{i,2}, \dots \mathbf{y}^{i,M}]$	Adaptive spectral-spatial tokens in one group
Local-Enhanced Transformer Encoder	$\mathbf{M}$	SNN attention mask
	$\mathbf{A}_{\text{snn}}$	SNN attention score
	$\mathbf{Y}_{L_1}$	Output features of local transformer encoder
	$\mathbf{Z}_{L_2}$	Output features of global transformer encoder
Classification Head	$\mathbf{F}$	Pixel-level features
	$\tilde{\mathbf{R}}$, $\mathbf{R}'$	Row-normalized association matrix, hard-association matrix
	$\mathbf{I}^{xy}$, $\bar{\mathbf{I}}^{xy}$	Original, reconstructed positional pixel features

generates low-level spectral–spatial tokens tailored for HSI and allows to explicitly model locality with the aid of a simple attention mask, taking full account of the image redundancy, 3-D property, and spatial prior knowledge of HSI.

III. LOCAL-ENHANCED SPECTRAL–SPATIAL TRANSFORMER

A. Overall Framework

Fig. 1 illustrates the architecture of LESSFormer and Table I summarizes the key notations in this article. Similar to ViT, LESSFormer consists of three modules: 1) HSI2Token Module; 2) local-enhanced transformer encoder; and 3) classification head. The HSI2Token module is first utilized to transform HSI into adaptive spectral–spatial tokens, which involves a shallow CNN, a spectral partition operation, and a spatial partition operation to minimize redundancy in HSI and yield discriminative low-level spectral–spatial representations. Subsequently, the output tokens are fed into the local-enhanced transformer encoder constructed by a novel local transformer encoder and a standard global transformer encoder in serial to further enhance the discriminative ability. By imposing a local spatial prior on the global SA module that restricts attention to a local neighborhood region, the local transformer encoder explicitly models locality to boost the classification performance when limited training samples are available, while the global transformer encoder is leveraged to preserve the capability to extract long-range information. Finally, the enhanced tokens are processed by a simple classification head module, which maps the enhanced token representations into the pixel-level feature maps and conducts classification to obtain the predicted labels for the whole image.

To optimize the parameters in LESSFormer, we employ the joint loss function that combines the cross-entropy loss, task-specific reconstruction loss, and compactness loss, that is

$$L = L_{\text{CE}}\left(\hat{\mathbf{G}}, \mathbf{G}\right) + w_1 L_{\text{recon}}(\mathbf{R}, \mathbf{G}) + w_2 L_{\text{cpt}}\left(\mathbf{R}, \mathbf{I}^{xy}\right) \quad (3)$$

where $\hat{\mathbf{G}}$ denotes the predict labels and $\mathbf{G}$ denotes the ground-truth, $\mathbf{I}^{xy}$ denotes the positional pixel features, and w_1 and w_2 are two weight coefficients. The $\mathbf{R}$ represents the pixel–superpixel association matrix, as detailed in Section III-B. The last two auxiliary losses are adopted to generate task-specific superpixels with low spatial variance [54], which can be, respectively, computed by

$$L_{\text{recon}}(\mathbf{R}, \mathbf{G}) = L_{\text{CE}}\left(\tilde{\mathbf{R}}\hat{\mathbf{R}}^T\mathbf{G}, \mathbf{G}\right) \quad (4)$$

$$L_{\text{cpt}}\left(\mathbf{R}, \mathbf{I}^{xy}\right) = \left\|\bar{\mathbf{I}}^{xy} - \mathbf{I}^{xy}\right\|_2 \quad (5)$$

where $\tilde{\mathbf{R}}$ is the row-normalized $\mathbf{R}$, $\hat{\mathbf{R}}$ is the column-normalized $\mathbf{R}$, $\tilde{\mathbf{R}}\hat{\mathbf{R}}^T\mathbf{G}$ represents the reconstructed labels, and $\bar{\mathbf{I}}^{xy}$ denotes the reconstructed positional pixel features, which can be determined by

$$\begin{aligned} \mathbf{S}^{xy} &= \hat{\mathbf{R}}^T\mathbf{I}^{xy} \\ \bar{\mathbf{I}}^{xy}_n &= \mathbf{S}^{xy}_m \mid \mathbf{R}'_n = m \end{aligned} \quad (6)$$

where $\mathbf{R}'_n = \underset{m\in\{1,\dots,M\}}{\arg\max}\, \mathbf{R}_{nm}$ is the hard-association matrix, and M is the superpixel number.

B. HSI2Token Module

The transformer is originally developed for NLP tasks and takes a sequence of tokens as input. In order to process images such as HSIs, ViT performs tokenization by splitting the raw image into a sequence of nonoverlapping patches of size $p \times p$ (e.g., 16×16) pixels, then applying flatten and linear projection operations to obtain token embeddings, as shown in Fig. 2(a). Note that there is an optional PCA step for dimension reduction with regard to HSIs. The original tokenization module lacks the ability to fully utilize the additional spectral information in HSI and preserve the low-level spatial features such as edges for HSI classification tasks.

To address these problems, we devise the HSI2Token module tailored for HSI, as illustrated in Fig. 2(b). The HSI2Token module consists of a shallow CNN, a group convolutional block, and a learnable superpixel segmentation submodule. Concretely, the original high-dimensional HSI data $\mathbf{H} \in \mathbb{R}^{H\times W\times B}$ is first fed into a two-layer CNN to minimize image redundancy and generate low-level discriminative features $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, where H, W, B, and C represent the height, width, band number, and the dimension of hidden features, respectively. Each convolutional block in the shallow CNN orderly involves a batch normalization (BN) layer, a 1×1 convolutional layer, and a rectified linear units (ReLU) activation function.

After that, we extract patches of the low-level features from the spectral dimension with a group convolutional block that

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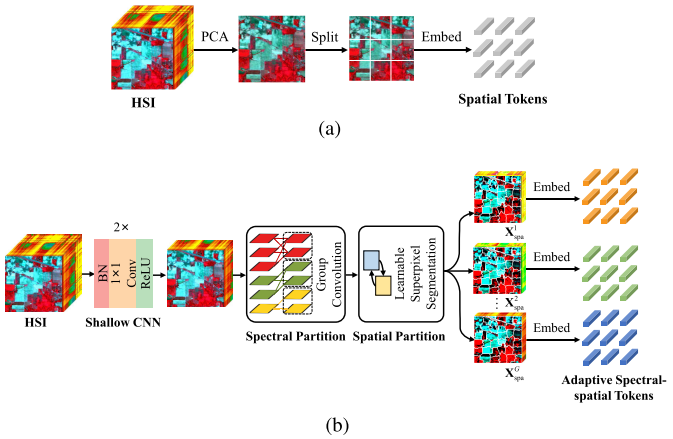


Fig. 2. Comparison of the original tokenization module and HSI2Token module. (a) Original tokenization module. (b) HSI2Token module.

contains the BN layer, 1×1 group convolutional layer, and ReLU activation function. The group convolutional block is an efficient alternative to the spectral partition operation, which divides the input channels and filters into G groups to make each group perform an independent convolution operation and generates a set of patches as follows:

$$\mathbf{X} \to \mathbf{X}_{\text{spe}} = \left[\mathbf{X}^1, \mathbf{X}^2, \ldots, \mathbf{X}^G\right] \tag{7}$$

where $\mathbf{X}_{\text{spe}}$ represents the set of patches generated by the group convolution operation, $\mathbf{X}^i \in \mathbb{R}^{H \times W \times C_g}$ denotes the spectral–spatial features of the ith group, $i = 1; 2; \ldots; G$, and C_g is the feature dimension of each group determined by $C_g = C/G$.

The differentiable Simple Linear Iterative Clustering (SLIC) submodule further divides the patch in each group into irregular subpatches along the superpixel boundary that closely sticks to the object boundary, that is,

$$\mathbf{X}^i \to \mathbf{X}^i_{\text{spa}} = \left[\mathbf{X}^{i,1}, \mathbf{X}^{i,2}, \ldots, \mathbf{X}^{i,M}\right] \tag{8}$$

where $\mathbf{X}^{i,j}$ is the features of jth irregular superpixel in the ith group, and $j = 1; 2; \ldots; M$. The superpixel number M is determined by $M = H \times W/\lambda$, where λ represents the segmentation scale.

Specifically, this submodule is employed to transform the pixel features (i.e., patches in each group) into the superpixel features (i.e., irregular subpatches) in a soft assignment manner with three steps [54]: First, initialize superpixel centers with the average pixel features of M regular grid cells. Second, alternately updates the pixel–superpixel associations and superpixel centers for v iterations to converge to a stable state. Finally, map pixel features into superpixel features with the final pixel–superpixel associations $\mathbf{R} \in \mathbb{R}^{M \times N}$, where N is the total pixel number. Then, the subpatches are flattened and projected into token embeddings $\mathbf{Y}^i = [\mathbf{y}^{i,1}, \mathbf{y}^{i,2}, \ldots, \mathbf{y}^{i,M}] \in \mathbb{R}^{M \times C_g}$ for each group, where $\mathbf{y}^{i,j} \in \mathbb{R}^{C_g}$ is the weighted feature vector of the jth superpixel in the ith group. In practice, we perform the spatial partition and embedding operations together through matrix multiplication, which is calculated in the following way:

$$\mathbf{Y}^i = \hat{\mathbf{R}}^T \overline{\mathbf{X}}^i \tag{9}$$

where $\overline{\mathbf{X}}^i \in \mathbb{R}^{N \times C_g}$ denotes the flattened $\mathbf{X}^i$ in (8).

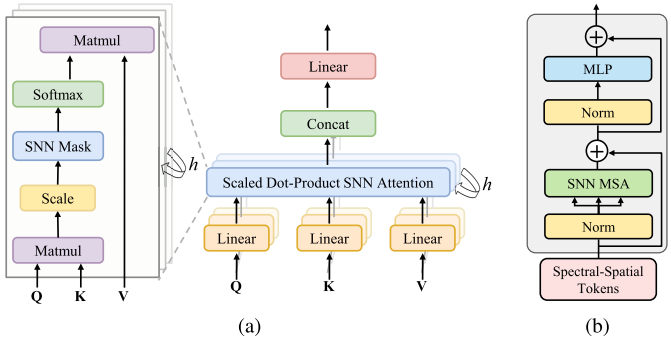


Fig. 3. Illustration of the SNN MSA layer and local transformer encoder. (a) SNN MSA layer. (b) Local transformer encoder.

It is worth noting that positional embedding is not adopted in our work since superpixels are unordered. Compared with the original tokenization module, the proposed HSI2Token module is capable of generating representative spectral–spatial tokens tailored for HSI data, with full accounting for image redundancy property and image-spectrum merged structure of the HSIs.

C. Local-Enhanced Transformer Encoder

Unlike the original transformer encoder, the novel local-enhanced transformer encoder consists of a local transformer encoder and a global transformer encoder stacked in serial. As the name indicates, the former is devised to capture the local spectral–spatial information among spatial neighboring tokens in each group, the latter is utilized to retain the ability to capture global relations among tokens from the spectral dimension.

1) Local Transformer Encoder: As detailed in Section II, the MSA layer plays a dominant role in building long-range relations between tokens among the crucial elements in the original transformer encoder. Nevertheless, the original global MSA layer overlooks the valuable local information in the spatial dimension, leading to performance degradation especially when limited training samples are available. Moreover, the original global MSA layer may take a few irrelevant tokens into consideration and introduce noise into the semantic modeling process [55], [56]. These limitations motivate us to devise the novel spatial nearest neighbor (SNN) MSA layer and construct the local transformer encoder based on it, as shown in Fig. 3(a) and (b).

Specifically, we impose the local context prior on the MSA layer by calculating attention values between spatial neighboring tokens to explicitly emphasize the local interactions between nearby land covers in HSI. In practice, in order to avoid misleading connections between the target token and the irrelevant tokens outside its neighborhood, we employ an SNN mask matrix $\mathbf{M}$ to element-wise multiply the attention matrix, which can be implemented inside the scaled dot-product attention by

$$\mathbf{M}_{pq} = \begin{cases} 1, & \text{if } p = q \text{ or } \mathrm{S}_p \text{ is adjacent to } \mathrm{S}_q \\ -\infty, & \text{otherwise} \end{cases}$$

$$\mathbf{A}_{\text{snn}} = \text{softmax}\left(\mathbf{M} \odot \frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right) \tag{10}$$

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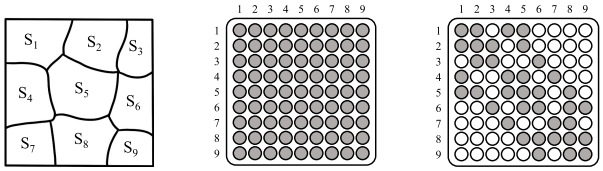


Fig. 4. Illustration of the SNN mask. (a) Superpixel segmentation map. (b) Attention mask of the MSA layer. (b) Attention mask of the SNN MSA layer. The white and gray colors represent the values of $-\infty$ and 1 in the attention mask matrices, respectively.

where p and q represent the row and column of $\mathbf{M}$, S_p and S_q denote the pth superpixel and qth superpixel, and $\mathbf{A}_{\text{snn}}$ denotes the SNN attention score matrix. Two superpixels are considered to be adjacent when they share a common boundary. It can be intuitively understood that the mask matrix of the original MSA layer is an all-ones matrix without restriction on the connections, while that of the SNN MSA layer is the SNN matrix $\mathbf{M}$ that reduces irrelevant connections, as illustrated in Fig. 4(b) and (c).

The local transformer encoder is devised to enhance the similarity among spatial neighboring tokens in each group based on the SNN MSA layer, that is,

$$\begin{aligned}\mathbf{Y}'^{i}_{l_1} &= \mathbf{Y}^{i}_{l_1-1} + \mathrm{MSA}_{\text{snn}}\big(\mathrm{LN}\big(\mathbf{Y}^{i}_{l_1-1}\big)\big)\\ \mathbf{Y}^{i}_{l_1} &= \mathbf{Y}'^{i}_{l_1} + \mathrm{MLP}\big(\mathrm{LN}\big(\mathbf{Y}'^{i}_{l_1}\big)\big)\end{aligned} \tag{11}$$

where $\mathbf{Y}^{i}_{l_1-1} = [\mathbf{y}^{i,1}_{l_1-1}, \mathbf{y}^{i,2}_{l_1-1}, \ldots, \mathbf{y}^{i,M}_{l_1-1}] \in \mathbb{R}^{M\times C_g}$ are the adaptive spectral–spatial tokens of layer (l_1-1), $l_1 = 1; 2; \ldots; L_1$, L_1 is the depth of the local transformer encoder, and $\mathrm{MSA}_{\text{snn}}$ denotes the SNN MSA layer. The input token of the first layer $\mathbf{Y}^{i}_{0}$ is the $\mathbf{Y}^{i}$ in (9).

2) Global Transformer Encoder: Considering that the original MSA layer and the SNN MSA layer supplement each other in modeling dependencies, we employ a standard global transformer encoder after the local transformer encoder to extract features from local to global. Concretely, tokens of all groups obtained by the local-enhanced transformer encoder are assembled into $\mathbf{Y}_{L_1} = [\mathbf{Y}^{1}_{L_1}, \mathbf{Y}^{2}_{L_1}, \ldots, \mathbf{Y}^{G}_{L_1}]$ and reshaped to $\mathbf{Z} \in \mathbb{R}^{M\times G\times C_g}$, then they are fed into the global transformer encoder. The global transformer encoder retains the ability to capture global relations among tokens from the spectral dimension as follows:

$$\begin{aligned}\mathbf{Z}'_{l_2} &= \mathbf{Z}_{l_2-1} + \mathrm{MSA}\big(\mathrm{LN}\big(\mathbf{Z}_{l_2-1}\big)\big)\\ \mathbf{Z}_{l_2} &= \mathbf{Z}'_{l_2} + \mathrm{MLP}\big(\mathrm{LN}\big(\mathbf{Z}'_{l_2}\big)\big)\end{aligned} \tag{12}$$

where $\mathbf{Z}_{l_2-1}$ are the token embeddings of layer (l_2-1), $l_2 = 1; 2; \ldots; L_2$, and L_2 is the depth of the global transformer encoder.

Although the global transformer encoder and the original transformer encoder have the same network architecture, there are still slight differences between them in building dependencies. The global transformer encoder are utilized to model long-range dependencies between tokens in different spectral groups, while the original transformer encoder captures long-range spatial relations between tokens in every group.

TABLE II
NUMBER OF TRAINING AND TEST DATA ON THE XA DATASET

No.	Color	Class name	Training	Test
1		Acer compound	30	49919
2		Willow	30	12786
3		Sophora japonica	30	52778
4		White wax	30	56333
5		Water	30	70467
6		Corn	30	28731
7		Pear tree	30	166134
8		Soybean	30	1273
9		Poplar	30	31455
10		Vegetable field	30	6016
11		Half bare land	30	1466
12		Grassland	30	10457
13		Peach	30	29844
14		House	30	14329
		Total	420	531988

D. Classification Head

Instead of adopting a single additional "classification token," we harness all the enhanced tokens to conduct classification, which can provide more discriminative information. For this purpose, we transform the sequence of enhanced spectral–spatial tokens $\mathbf{Z}_{L_2}$ into the image-like feature representations $\mathbf{F}$ with the row-normalized pixel–superpixel associations $\tilde{\mathbf{R}}$, which can be computed by simple matrix multiplication as follows:

$$\mathbf{F} = \tilde{\mathbf{R}}\mathbf{Z}_{L_2} \tag{13}$$

Then the output features from each group are concatenated and fed into the softmax classifier that contains a linear projection layer and a softmax activation layer to predict the class probabilities of all pixels in HSI.

IV. EXPERIMENTS

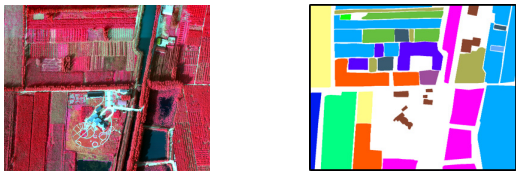
A. Data Description

The classification performance of the LESSFormer method is evaluated on three benchmark HSI datasets, including the XA, PU, and HU. The details of the three datasets are described as follows.

1) Xiong'an (XA): The XA dataset[1] was obtained by the advanced hyperspectral imager (AHSI) sensor over Hebei, China, which is composed of 1580 × 3750 pixels with 0.5 m per pixel resolution and has 256 spectral bands. A cropped image of size 800 × 1000 pixels was selected from this dataset for experiments, which preserves abundant land cover classes (i.e., 14 categories). Table II shows the category information as well as the amounts of training and test data for each class. The false color image and ground truth map of the XA dataset are displayed in Fig. 5.

[1] http://www.hrs-cas.com/a/share/shujuchanpin/2019/0501/1049.html

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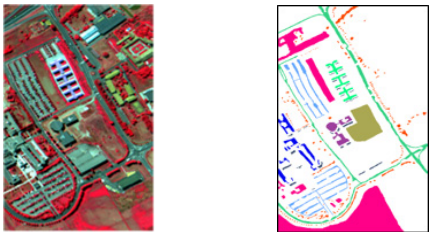


(a) (b)

Fig. 5. XA dataset. (a) False color image. (b) Ground truth map.

TABLE III
NUMBER OF TRAINING AND TEST DATA ON THE PU DATASET

No.	Color	Class name	Training	Test
1		Asphalt	30	6601
2		Meadows	30	18619
3		Gravel	30	2069
4		Trees	30	3034
5		Painted metal sheets	30	1315
6		Bare Soil	30	4999
7		Bitumen	30	1300
8		Self-Blocking Bricks	30	3652
9		Shadows	30	917
		Total	270	42506



(a) (b)

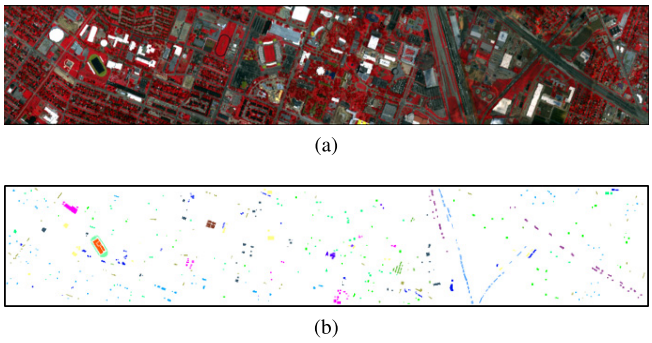
Fig. 6. PU dataset. (a) False color image. (b) Ground truth map.

2) Pavia University (PU): The reflective optics system imaging spectrometer (ROSIS) sensor gathered the data over the city of Pavia, Italy at 1.3-meter pixels. This image contains 610 × 340 pixels and 103 spectral bands. The ground truth comprises nine classes, such as "Asphalt," "Meadows," and "Trees." The category information and the number of training and test samples for each class are detailed in Table III. The false color image and ground truth map of the PU dataset are shown in Fig. 6.

3) Houston University (HU): The compact airborne spectrographic imager (CASI) sensor collected the data over the University of Houston campus and its neighboring urban area, US at 2.5-meter pixels. The area comprises 1905 × 349 pixels and 144 spectral channels. 15 classes depicting different land cover types are defined. This dataset was distributed in the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest but processed into a cloud-free version. The category information and the number of training and test pixels for each class are listed in Table IV. The false color image and ground truth map of the HU dataset are displayed in Fig. 7.

TABLE IV
NUMBER OF TRAINING AND TEST DATA ON THE HU DATASET

No.	Color	Class name	Training	Test
1		Grass Healthy	198	1053
2		Grass Stressed	190	1064
3		Grass Synthetic	192	505
4		Tree	188	1056
5		Soil	186	1056
6		Water	182	143
7		Residential	196	1072
8		Commercial	191	1053
9		Road	193	1059
10		Highway	191	1036
11		Railway	181	1054
12		Parking Lot1	192	1041
13		Parking Lot2	184	285
14		Tennis Court	181	247
15		Running Track	187	473
		Total	2832	12197



(a) (b)

Fig. 7. HU dataset. (a) False color image. (b) Ground truth map.

TABLE V
CONFIGURATIONS OF LESSFORMER FOR XA, PU, AND HU DATASETS. THERE IS A SLIGHT DIFFERENCE IN THE HYPERPARAMETERS ON THREE DATASETS

Module	Layer	Configuration
HSI2Token Module	conv 1-1	1×1@128
	conv 1-2	1×1@128
	gconv 1-1	1×1@128, G=2/4/4
Local-Enhanced Transformer Encoder	local transformer encoder	L_1=3/2/1, h=1/4/1
	global transformer encoder	L_2=1/2/3, h=1/4/1
Classification Head	fc	@#Classes

B. Experiment Setup

1) Implementation Details: The network configuration of the LESSFormer method is detailed in Table V, which varies

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TABLE VI
ACCURACY COMPARISONS ON THE XA DATASET

ID	Traditional method		CNN-based/FCN-based method			ViT-based method				
	SVM	EMP-SVM	HPDM-SPRN	MSRN	FreeNet	ViT	HiT	SpectraFormer	SSFTT	LESSFormer
1	74.98±2.8	86.6±2.28	93.02±2.61	**95.72±1.91**	94.79±3.65	84.74±2.92	65.82±14.1	77.93±4.89	90.16±1.46	95.05±2.86
2	81.78±2.77	93.23±2	99.62±0.75	92.04±3.83	**100±0**	95.5±4.21	73.38±2.2	73.38±8.57	99.18±0.19	99.86±0.29
3	72.97±3.26	77.62±3.68	87.28±3.88	**94.42±2.37**	93.05±0.51	77.44±2.91	53.92±2.43	51.58±5.87	90±2.28	93.42±3.34
4	90.4±2.14	94.53±2.77	**99.47±0.53**	96.79±2.25	98.74±0.86	94.88±1.85	90.29±1.31	85.99±7.96	97.31±2.06	98.87±0.86
5	84.67±3.32	83.71±3.88	93.69±1.71	**98.08±1.11**	95.06±1.45	81.6±1.41	67.58±1.14	77.13±4.11	90.98±5.4	96.47±1.27
6	70.69±6.13	67.91±3.66	95.16±3.46	83.96±4.52	93.71±0.43	69.79±4.29	53.79±6.52	60.32±2.59	92.06±0.43	**95.41±2.38**
7	51.05±4.23	64.04±1.23	75.81±2.82	**98.52±0.33**	71.93±1.1	53.62±7.1	56.49±3.68	42.83±3.28	74.38±1.64	78.19±2.37
8	89.76±1.54	93.78±2.25	**99.61±0.53**	82.85±1.93	99.37±0.59	97.17±1.56	76.62±19.37	89.05±2.51	99.05±1.2	99.09±0.9
9	82.62±4.29	78.3±7.56	93.33±2.7	83.03±3.95	91.68±5.63	83.94±3.49	34.23±5.06	59.08±6.92	**95.13±1.69**	89.16±5.06
10	61.1±5.13	64.78±5.5	93.6±3	34.86±5.16	**97.87±1.55**	74.48±12.07	33.06±4.32	45.87±8.81	91.95±5.11	95.19±2.2
11	78.79±5.84	88.28±3.99	98.02±2.6	20.57±7.96	**100±0**	92.91±2.61	36.47±20.63	65.87±12.7	97.08±2.44	99.24±0.22
12	56.39±3.63	70.68±2.05	91±3.38	58.46±5.94	93.04±0.29	61.47±10.42	35.37±7.48	52.58±11.87	86.94±3.44	**94.56±2.69**
13	38.94±5.89	56.56±9.79	89.2±4.07	82.43±7.61	92.87±2.69	66.73±6.74	30.66±18.99	61.88±5.92	78.01±6.91	**93.74±2.19**
14	91.29±1.76	89±1.26	94.68±1.26	92.04±4.03	96.38±0.87	86.94±8.17	85.11±1.15	87.42±4.04	92.99±1.04	**96.96±1.47**
OA	68.54±2.05	75.62±1.49	87.97±0.69	89.74±0.29	87.78±0.35	72.89±1.24	59.74±1.64	61.42±1.96	86.09±1.00	**89.99±0.40**
AA	73.24±1.31	79.22±1.11	93.11±0.40	79.56±0.50	94.18±0.18	80.09±1.28	56.84±3.29	66.49±2.90	91.09±0.96	**94.66±0.42**
Kappa	64.39±2.23	71.96±1.71	86.09±0.76	88.09±0.33	85.90±0.38	69.25±1.20	53.99±1.63	56.69±2.05	83.91±1.14	**88.39±0.45**

TABLE VII
ACCURACY COMPARISONS ON THE PU DATASET

ID	Traditional method		CNN-based/FCN-based method			ViT-based method				
	SVM	EMP-SVM	HPDM-SPRN	MSRN	FreeNet	ViT	HiT	SpectraFormer	SSFTT	LESSFormer
1	73.85±3.59	93.4±0.8	92.87±5.86	**98.38±1.08**	97.88±2.44	68.21±10.22	74.71±7.01	73.81±2.15	90.16±1.46	98.29±0.84
2	76.71±5.87	82.7±7.38	96.55±2.51	**99.65±0.27**	97.48±1.39	81.12±5.13	87.79±3.82	90.52±0.99	94.76±5.86	97.54±1.46
3	75.99±4.12	88.62±3.68	92.44±6.26	97.4±1.6	**99.63±0.36**	87.46±0.31	70.02±12.88	64.04±4.72	89.57±4.8	99.62±0.39
4	89.55±2.49	95.81±2.25	94.09±3.23	90.91±4.28	**97.24±1.42**	81.48±3.68	92.38±1.47	91.25±6.69	91.88±1.03	97.13±1.04
5	99.21±0.1	99.65±0.04	**100±0**	96.99±2.48	99.95±0.09	92.78±8.07	99.39±0.46	99.92±0.11	99.63±0.2	99.7±0.19
6	82.23±7.31	80.25±6.65	99.46±0.55	92.46±6.58	99.44±0.94	88.66±1.92	62.7±12.21	60.36±5.13	95.81±1.8	**99.88±0.02**
7	89.69±4.06	95.22±0.71	**100±0**	97.73±2.25	99.64±0.62	97.08±1.74	80.69±4.56	84.12±0.27	98.52±1.2	99.86±0.09
8	80.59±5.68	88.87±4.9	94.5±5.38	92.06±2.91	96.89±0.25	82.57±2.93	79.06±4.9	80.72±8.09	87.25±3.61	**98.94±1.05**
9	99.85±0.09	**99.94±0.05**	97.62±1.38	91.39±3.66	99.38±0.56	75.96±1.93	96.11±2.48	98.26±2.31	96.21±3.46	99.26±0.37
OA	79.72±1.78	87.11±2.50	96.00±2.00	96.69±1.02	97.99±0.25	81.20±0.18	81.84±1.46	82.56±1.16	93.08±2.80	**98.20±0.73**
AA	85.30±3.70	91.61±2.94	96.39±1.49	95.22±0.69	98.62±0.30	83.93±1.70	82.54±1.80	82.56±1.17	93.55±0.83	**98.91±0.28**
Kappa	74.10±1.98	83.33±2.98	94.74±2.59	95.63±1.32	97.35±0.32	75.93±0.01	76.06±1.64	76.91±1.47	90.93±3.55	**97.76±0.96**

slightly on three datasets. In terms of the optimization hyperparameters, the epochs are set as 500, 300, and 500 for the XA dataset, PU dataset, and HU dataset, respectively. We adopt the Adam optimizer with a learning rate equal to 0.0005 without learning rate decay. The batch size is set to 1 since we process the whole image at once. In addition, we set the iteration number v to 5. Our experiments have been conducted on an NVIDIA server with a Titan RTX 3090 graphics processing unit (GPU). The software environment comprises Python-3.8.5 and PyTorch-1.1.0. As to the training and test sets, two sampling strategies are adopted to perform a comprehensive evaluation. Specifically, for the XA and PU datasets, 30 pixels from labeled samples in each class are randomly chosen to form the training sets and the other pixels are used to form the test sets. While the fixed disjoint training-test set is exploited for the HU dataset. Each experiment is repeated five times. The overall accuracy (OA), average accuracy (AA), kappa coefficient, and per-class accuracy (CA) are adopted as the evaluation metrics.

2) Comparison Methods: In order to thoroughly validate the effectiveness of LESSFormer, nine HSI classification methods are chosen for comparison, including two traditional methods, i.e., SVM and extended morphological profile (EMP)-SVM, two CNN-based methods, i.e., homogeneous pixel detection module (HPDM)-spectral partitioning residual network (SPRN) [12] (Spectral Partitioning Residual Network), and

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TABLE VIII
ACCURACY COMPARISONS ON THE HU DATASET

ID	Traditional method		CNN-based/FCN-based method			ViT-based method				
	SVM	EMP-SVM	HPDM-SPRN	MSRN	FreeNet	ViT	HiT	SpectraFormer	SSFTT	LESSFormer
1	83.57	82.15	78.27±0.64	**100±0**	81.45±0.95	79.54±1.69	81.44±1.06	83.82±1.48	90.16±1.46	81.31±0.31
2	98.21	**98.87**	90.28±2.88	88.17±4.18	92.36±6.24	91.92±3.88	95.45±2.58	98.46±0.84	94.76±5.86	98.67±1.43
3	99.8	99.6	97.27±0.15	98.39±2.27	98.28±1.13	92.91±2.8	89.45±9.59	87.33±8.41	94.72±3.54	**100±0**
4	98.3	91	90.34±3.32	95.98±1.21	98.67±0.98	93.63±1.27	92.45±1.36	97.98±0.3	89.93±3.94	**99.26±0.18**
5	98.77	99.43	99.66±0.32	99.53±0.81	**100±0**	97.16±2.12	91.67±1.2	99.65±0.38	99.74±0.28	99.26±0.42
6	97.9	**99.3**	95.24±0.69	69.69±13	91.84±9.7	81.96±1.74	89.15±1.02	90.68±3.3	97.44±2.25	95.24±0.52
7	87.13	**92.91**	72.22±2.07	91.85±1.02	90.95±2.03	81.72±4.5	90.63±0.51	86.38±4.12	78.39±7.06	87.85±3.06
8	71.32	81.2	**87.45±4.45**	81.81±6.92	78.76±3.47	82.28±5.21	69.88±1.99	81.45±4.27	62.78±8.54	84.92±3.82
9	66.01	76.2	83.59±2.27	81.65±8.15	89.42±4.01	84.1±0.79	78.4±7.59	77.34±1.39	**89.77±2.67**	85.1±3.12
10	89.67	67.09	88.05±4.36	**92.52±4.03**	75.16±11.17	67.53±8.55	71.18±5.13	76.13±15.42	64.77±4.8	75.66±14.66
11	84.73	96.58	**98.25±0.96**	78.52±9.96	95.41±2.61	91.56±4.16	79.97±4.15	76.6±6.23	91.52±5.04	93.4±5.85
12	61.77	72.24	91.91±1.87	**93.69±1.94**	90.46±8.51	85.88±1.16	82.79±2.33	81.08±3.35	90.36±1.96	89.76±4.78
13	62.81	72.28	72.07±2.65	**94.83±3.9**	78.13±6.28	75.93±3.59	85.73±3.7	71.7±3.78	91.7±2.03	72.21±9.36
14	99.19	**100**	**100±0**	99.6±0.69	99.6±0.4	99.19±0.95	90.37±2.41	92.85±6.46	**100±0**	**100±0**
15	97.25	**100**	99.92±0.17	92.67±1.5	87.88±12.67	69.9±21.54	80.21±1.69	96.55±2.84	99.22±0.8	99.96±0.08
OA	85.13	87.09	88.78±0.61	89.77±0.67	89.59±0.92	85.27±1.72	83.94±0.42	86.26±1.96	85.67±0.46	**90.26±1.40**
AA	86.43	88.59	89.63±0.42	90.59±1.29	89.89±0.77	85.01±1.98	84.58±0.73	86.53±2.54	88.16±0.29	**90.84±1.39**
Kappa	83.85	85.98	87.83±0.65	88.90±0.73	88.70±1.00	84.01±1.87	82.63±0.46	85.08±2.12	84.50±0.50	**89.42±1.53**

MSRN [57] (Multiscale Residual Network), one FCN-based method, i.e., FreeNet [28] (Patch-free Network), as well as four transformer-based methods, i.e., ViT [40], HiT [58] (HSI Transformer), SpectralFormer [47] (Spectral Transformer) and SSFTT [49] (Spectral-Spatial Feature Tokenization Transformer).

For the SVM, the radial basis function (RBF) kernel is selected as the default, and the fivefold cross validation is utilized to search for the optimal hyperparameters γ in the range of {0.1, 0.01, 0.001} and C in the range of {1, 10, 100, 1000}. For the EMP-SVM, we first extract the structural features of HSI by carrying out several openings and closings (i.e., 13, 7, 21 for the XA, PU, and HU datasets, respectively) on the first four principal components, where the disk starts from one-pixel size and expands one pixel each step, then SVM is used for classification. For the DL-based classification methods, we adopt the default values from open source codes and further carefully tuned the hyperparameters to obtain optimal performance.

C. Classification Results

We evaluated the classification performance of various comparison methods on the XA, PU, and HU datasets quantitatively and qualitatively.

1) Quantitative Evaluation: Tables VI–VIII report the mean and the standard deviation of each criterion index obtained by comparison methods on three datasets, where the highest value is highlighted in bold and the second highest value is indicated with an underscore. In terms of the traditional methods, SVM and EMP-SVM are inferior to most DL methods, since they fail to automatically extract the inherent spectral–spatial features in HSI. Considering the DL methods, the CNN-based and FCN-based methods generally achieve superior accuracies on three datasets owing to their outstanding ability to capture contextual information, especially local contextual relations. Despite the fact that the vision transformers are able to capture the long-range spatial features and spectral features, the transformer-based comparison methods (i.e., the ViT, HiT, SpectralFormer, and SSFTT methods) still yield unsatisfactory performance. Mainly because these methods overlook the image redundancy as well as the 3-D attributes of HSI data and have difficulty in implicitly modeling locality given insufficient training samples. On the contrary, the proposed LESSFormer method shows a significant superiority over the other transformer-based classification methods by generating low-level spectral–spatial tokens tailored for HSI and capturing both the local and long-range dependencies.

TABLE IX
OA (%) INDICES OF THE COMBINATIONS OF DIFFERENT SUBMODULES IN THE HSI2TOKEN MODULE ON THREE DATASETS

Baseline 1	Shallow CNN	Spectral Partition	Spatial Partition	OA		
				XA	PU	HU
✓	-	-	-	78.84±1.71	83.86±2.16	68.54±0.38
✓	✓	-	-	87.78±1.42	88.87±1.34	78.22±0.80
✓	✓	✓	-	88.25±0.73	90.30±0.92	81.21±2.40
✓	✓	✓	✓	**89.99±0.40**	**98.20±0.73**	**90.26±1.40**

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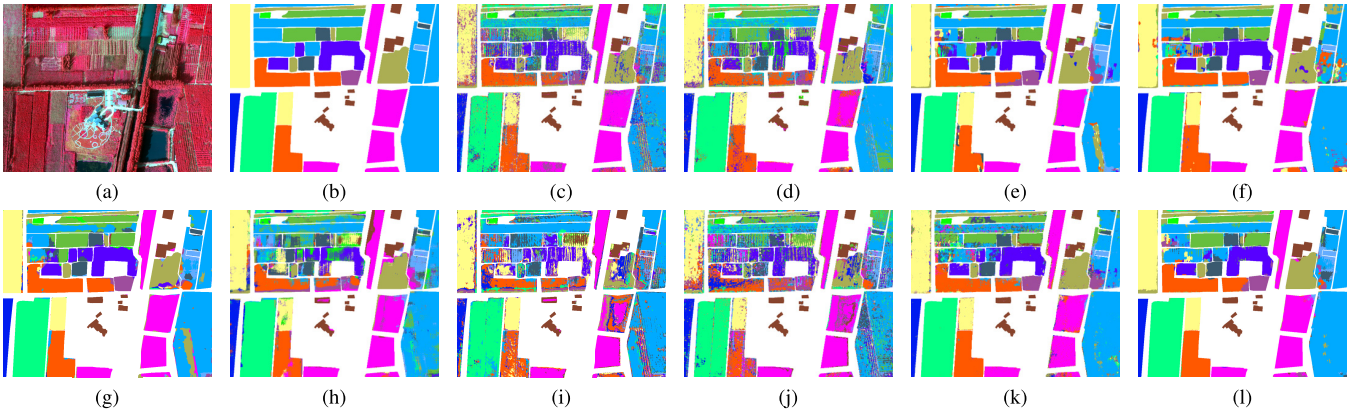


Fig. 8. Classification maps generated by different methods on the XA dataset. (a) False color image. (b) Ground truth map. (c) SVM (OA = 68.54%). (d) EMP-SVM (OA = 75.62%). (e) HPDM-SPRN (OA = 87.97%). (f) MSRN (OA = 89.74%). (g) FreeNet (OA = 87.78%). (h) ViT (OA = 72.89%). (i) HiT (OA = 59.74%). (j) SpectralFormer (OA = 61.42%). (k) SSFTT (OA = 86.09%). (l) LESSFormer (OA = 89.99%).

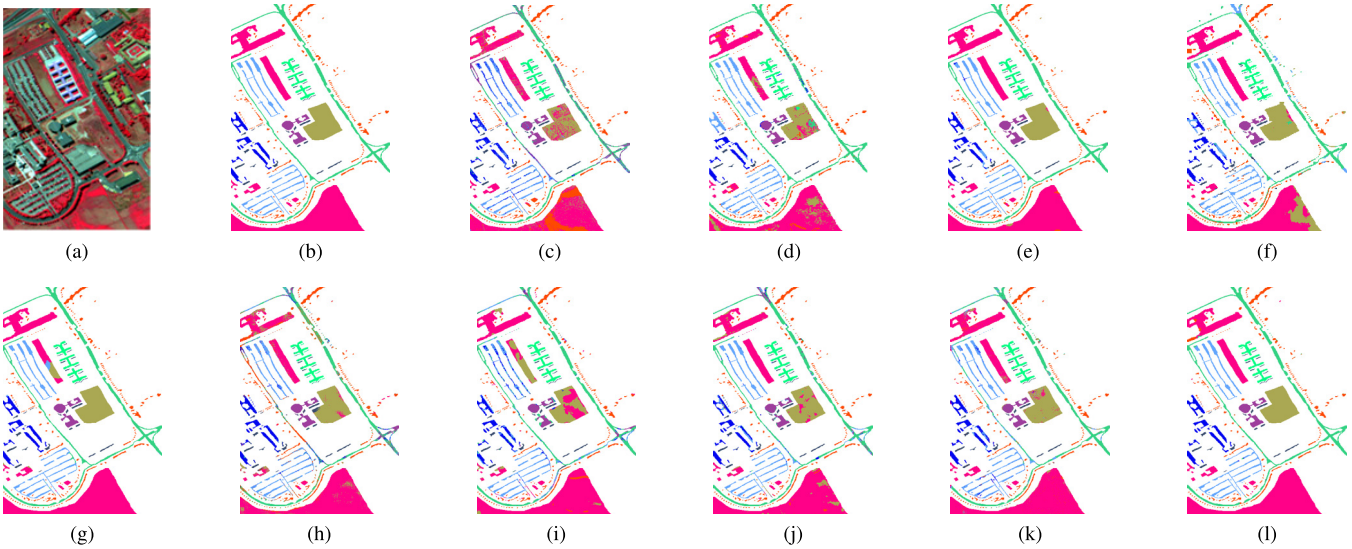


Fig. 9. Classification maps generated by different methods on the PU dataset. (a) False color image. (b) Ground truth map. (c) SVM (OA = 79.72%). (d) EMP-SVM (OA = 87.11%). (e) HPDM-SPRN (OA = 96.00%). (f) MSRN (OA = 96.69%). (g) FreeNet (OA = 97.99%). (h) ViT (OA = 81.20%). (i) HiT (OA = 81.84%). (j) SpectralFormer (OA = 82.56%). (k) SSFTT (OA = 93.08%). (l) LESSFormer (OA = 98.20%).

Overall, the proposed LESSFormer method surpasses all comparison methods in classification accuracies for three datasets. Compared to the second best model, LESSFormer increases the classification accuracy by 0.25% of OA, 0.48% of AA, and 0.30% of Kappa coefficient for XA dataset, by 0.21% of OA, 0.29% of AA, and 0.41% of Kappa coefficient for PU dataset, and by 0.49% of OA, 0.25% of AA, and 0.52% of Kappa coefficient for HU dataset. In addition, LESSFormer yields higher classification accuracies in most categories for three datasets. For instance, LESSFormer is quite competent at distinguishing different crops and tree species with similar reflectance patterns on the XA dataset, the highest CA can be found in four classes (i.e., Corn, Grassland, Peach, and House) and the second highest CA can be noticed in eight classes. On the PU dataset, LESSFormer outperforms the other methods in two classes and achieves the second best performance in five classes. Similarly, for HU dataset, the best accuracies are found in three classes.

2) Qualitative Evaluation: Figs. 8, 9, and 10 present the classification maps produced by the comparative methods on three datasets. For XA dataset, the classification map of LESSFormer shows more consistent agreement with ground truth, especially in the upper-left region with various similar crops and tree species. Similarly, for PU dataset, there are less noisy scattered points and misclassification in the classification map obtained from LESSFormer, while the other methods exhibit serious misclassification in some classes such as "Meadows." For HU dataset, the classification map generated by LESSFormer is relatively noise-free and preserves the detailed information well. Moreover, a subset extracted from the original image is outlined and enlarged for better delineation. It can be seen that a few methods misclassify the "Commercial" and "Highway" classes, due to the high spectral variance. In contrast, LESSFormer alleviates the problem by simultaneously exploiting the diagnostic spectral information and effective spatial information.

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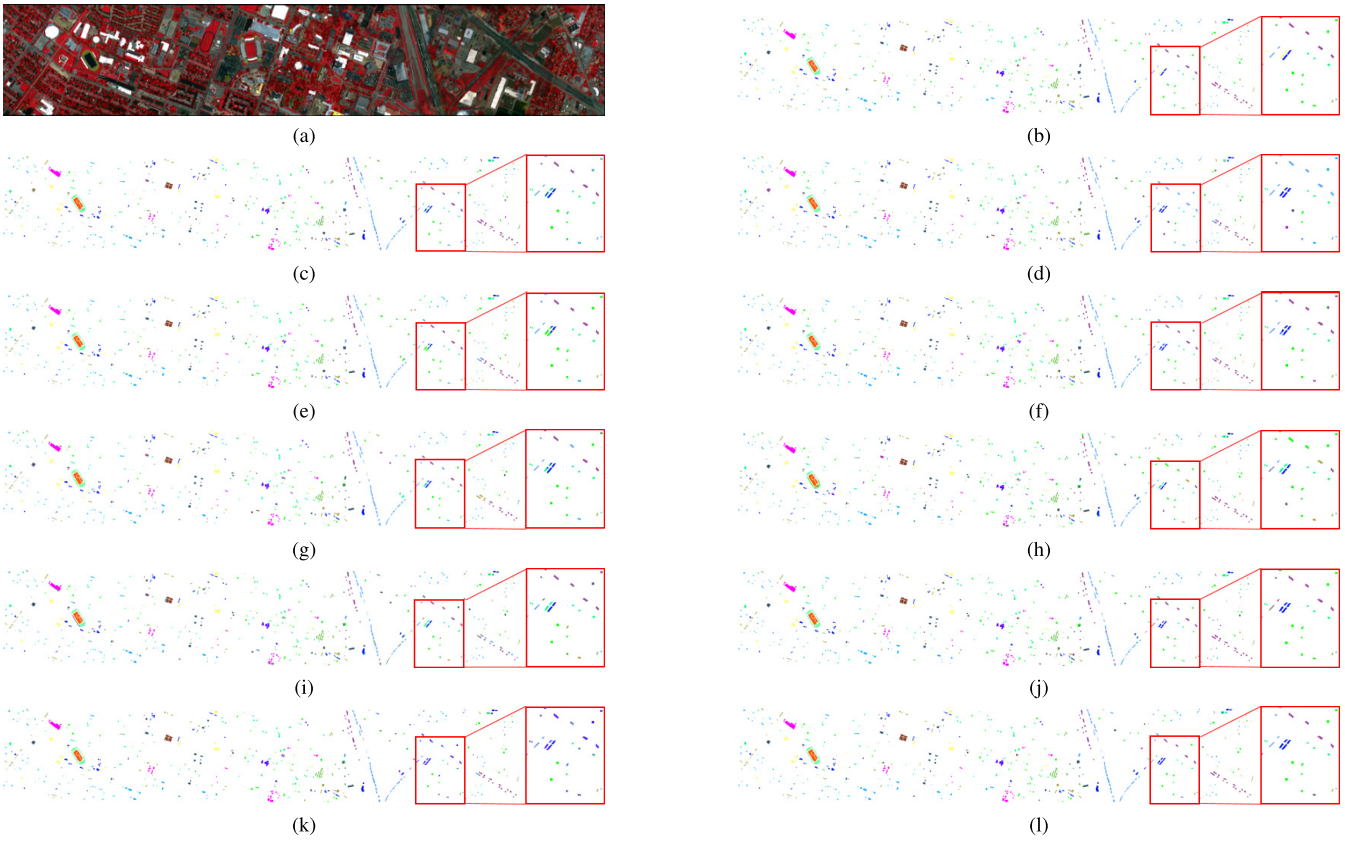


Fig. 10. Classification maps generated by different methods on the HU dataset. (a) False color image. (b) Ground truth map. (c) SVM (OA = 85.13%). (d) EMP-SVM (OA = 87.09%). (e) HPDM-SPRN (OA = 88.78%). (f) MSRN (OA = 89.77%). (g) FreeNet (OA = 89.59%). (h) ViT (OA = 85.27%). (i) HiT (OA = 83.94%). (j) SpectralFormer (OA = 86.26%). (k) SSFTT (OA = 85.67%). (l) LESSFormer (OA = 90.26%).

D. Discussion

1) Ablation Study: To further highlight the effectiveness of the key modules in LESSFormer (i.e., the HSI2Token module and local-enhanced transformer encoder), we carried out several ablation studies on three datasets. Tables IX and X present the classification performance with the combination of different submodules in the HSI2Token module and the local-enhanced transformer encoder, where the highest value is highlighted in bold and the second highest value is indicated with an underscore. The "baseline 1" method denotes the model that applies the original tokenization module, and the "baseline 2" method denotes the model that removes the transformer encoder.

Table IX shows that the shallow CNN increases accuracy significantly by 8.94% of OA on the XA dataset, by 5.01% of OA on the PU dataset, and by 9.68% of OA on the HU dataset, since this submodule can minimize image redundancy and generate low-level features, leading to more accurate token representations. Furthermore, the spectral partition and spatial partition also conduce to yield higher performance on all three datasets, where the spatial partition submodule makes a substantial contribution by preserving more detailed spatial information. According to Table X, the "baseline 2" method inevitably provides poor performance than the ones with a transformer encoder, since the transformer encoder can significantly promote the discriminant ability of token embeddings. In addition, both the local transformer encoder and the global transformer contribute to the gain of classification accuracy, which complement each other. Specifically, the local transformer encoder leads to higher performance by 3.91% of OA on the XA dataset, by 2.23% of OA on the PU dataset, and by 7.20% of OA on the HU dataset, and the global transformer encoder improves accuracy by 1.53% of OA on the XA dataset and by 6.99% of OA on the HU dataset. With the combination of the two kinds of transformer encoders, the model delivers the most accurate classification performance.

2) Parameter Sensitivity Analysis: The LESSFormer method involves four hyperparameters derived from two modules and two hyperparameters derived from the joint loss function.

1) HSI2Token Module: G (number of spectral groups), λ (segmentation scale).
2) Local-enhanced Transformer Encoder: L_1 (number of layers in the local transformer encoder), L_2 (number of layers in the global transformer encoder).
3) Joint Loss Function: ω_1 (weight coefficient of reconstruction loss), ω_2 (weight coefficient of compactness loss).

In this section, the parameter sensitivity of LESSFormer regarding different values of the four hyperparameters is explored. Fig. 11 presents the average value of OA indices when varying the number of spectral groups in the range of

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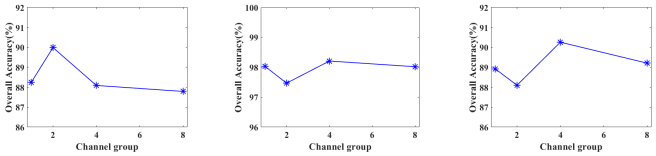


Fig. 11. OA (%) indices of LESSFormer with different number of groups for three datasets. (a) XA. (b) PU. (c) HU.

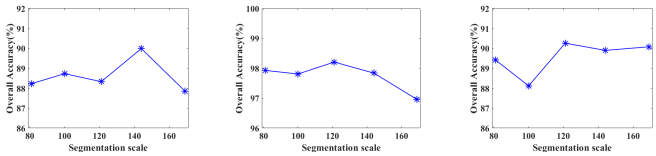


Fig. 12. OA (%) indices of LESSFormer with different segmentation scales for three datasets. (a) XA. (b) PU. (c) HU.

{2, 4, 8}, it is observed that the performance of LESSFormer is relatively stable with different number of groups. LESSFormer yields the highest accuracy when G is set to two for XA dataset, and four for PU as well as HU datasets. Fig. 12 displays the average value of OA indices with different segmentation scales in the range of {81, 100, 121, 144, 169} for three datasets. The highest accuracies are achieved when λ is set to 144 on the XA dataset, and 121 on the PU as well as HU datasets. A smaller λ means more superpixel numbers and less spatial contextual information contained in one superpixel. In contrast, a larger λ indicates fewer superpixel numbers, which will lead to the "mixed superpixels" issue that one superpixel includes pixels of different categories.

In terms of the number of layers L_1 and L_2, we set them in the range of {1+1, 1+2, 1+3, 2+1, 2+2, 2+3, 3+1, 3+2, 3+3} and report the accuracy on three datasets. According to Fig. 13, the optimal numbers of layers of the local transformer encoder and the global transformer encoder vary a lot for the three datasets. This might have occurred because the impact of local and long-range features is different in various HSI datasets. Considering that stacking more layers may lead to higher accuracy but higher computational cost, we choose the combination of "3+1" for XA dataset, "2+2" for PU dataset, and "1+3" for Houston dataset with tradeoffs between accuracy and efficiency.

Lastly, we set the weight coefficients w_1 and w_2 in the range of {0, 0.001, 0.01, 0.1} on three datasets and display the classification results in Fig. 14. The results indicate that the weight coefficients have a slight impact on the classification performance, and we set the weight coefficients w_1 and w_2 to 0.01 and 0.001 due to their high performance on three datasets.

3) Impact of the Number of Training Samples: Collecting large amounts of training samples is impractical and infeasible in real-world scenarios, thus the small sample classification capacity is necessary when classification methods are applied to practice. In order to investigate the performance of comparison methods and the proposed LESSFormer method when

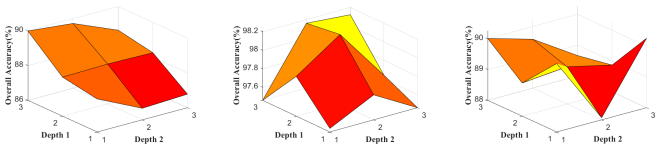


Fig. 13. OA (%) indices of LESSFormer with different network depth for three datasets. (a) XA. (b) PU. (c) HU.

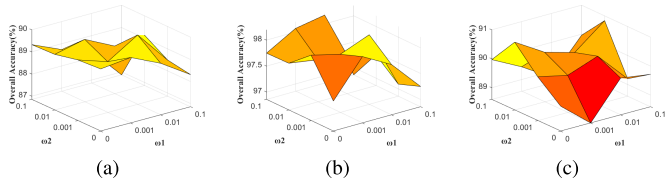


Fig. 14. OA (%) indices of LESSFormer with different weight coefficients of the joint loss function for three datasets. (a) XA. (b) PU. (c) HU.

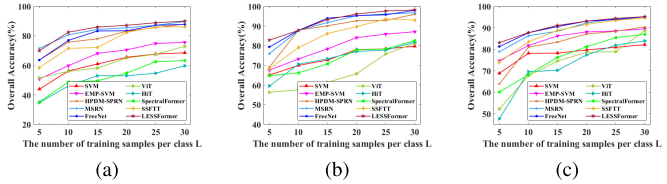


Fig. 15. OA (%) indices of LESSFormer with different number of training samples for three datasets. (a) XA. (b) PU. (c) HU.

limited training samples are available, we randomly choose different numbers of pixels in the range of {5, 10, 15, 20, 25, 30} on three datasets for classification. The OA indices of classification methods with different numbers of training samples are displayed in Fig. 15. As expected, the classification methods generally yield higher accuracies as the training sample size increases. Furthermore, we notice that there is a relatively sharp rise in accuracy for the ViT and SpectralFormer methods with more training samples, mainly because transformer-based models generally need large amounts of training data for decent accuracy. In contrast, LESSFormer consistently delivers higher or comparative performance than the other comparison methods on three datasets, illustrating the impressive small sample classification ability of our model.

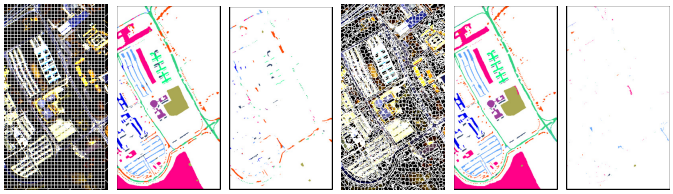
4) Visualization Analysis: To intuitively demonstrate the effectiveness of the essential components in LESSFormer, the segmentation maps obtained by the original tokenization module and the HSI2Token module as well as the corresponding classification results are presented in Fig. 16, where the first three pictures represent the segmentation map, classification map, and classification difference map acquired using the original tokenization module. The last three pictures display the corresponding results adopting the HSI2Token module. It can be observed from Fig. 16 that the HSI2Token module performs spatial partition along the superpixel boundary that fit the irregular objects boundary closely, leading to less misclassification in the boundary area of the classification results.

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TABLE X

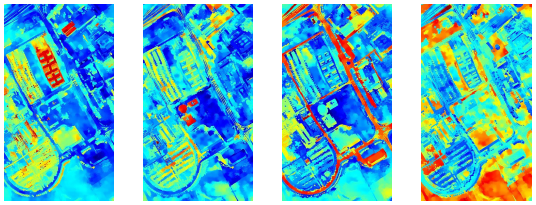
OA (%) INDICES OF THE COMBINATIONS OF DIFFERENT SUBMODULES IN THE LOCAL-ENHANCED TRANSFORMER ENCODER ON THREE DATASETS

Baseline 2	Original Transformer Encoder	Local Transformer Encoder	Global Transformer Encoder	Local-Enhanced Transformer Encoder	OA		
					XA	PU	HU
✓	-	-	-	-	78.32±1.64	88.00±1.69	77.46±4.64
✓	✓	-	-	-	85.27±0.74	95.33±1.54	82.87±1.99
✓	-	✓	-	-	89.18±1.00	97.56±1.33	90.07±0.52
✓	-	-	✓	-	86.80±1.07	95.20±0.87	89.86±2.14
✓	-	-	-	✓	**89.99±0.40**	**98.20±0.73**	**90.26±1.40**



(a) (b) (c) (d) (e) (f)

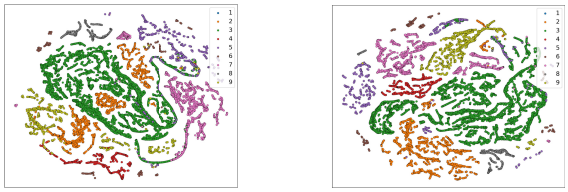
Fig. 16. Demonstration on the impact of the HSI2Token module on the PU dataset. (a) Segmentation result using the original tokenization module. (b) Classification map using the original tokenization module. (c) Classification difference map using the original tokenization module. (d) Segmentation result using the HSI2Token module. (e) Classification map using the HSI2Token module. (f) Classification difference map using the HSI2Token module.



(a) (b) (c) (d)

Fig. 17. Visualization of the typical feature maps obtained by the local transformer encoder and global transformer encoder on the PU dataset. (a) First typical feature map of the local transformer encoder. (b) Second typical feature map of the local transformer encoder. (c) First typical feature map of the global transformer encoder. (d) Second typical feature map of the global transformer encoder.

In order to intuitively illustrate the local information and the long-range information extracted by LESSFormer, several typical feature maps obtained by the local transformer encoder and global transformer encoder on the PU dataset are displayed in Fig. 17, where the first two pictures show the output feature maps of local transformer encoder and the last two pictures display the corresponding results of global transformer encoder. It is observed that the highly activated image regions (i.e., the red parts) in the feature maps of local transformer encoder are relatively concentrated, which is especially useful for distinguishing the small-scale objects such as "Painted metal sheets" and "Bitumen." On the contrary, the feature maps of the global transformer encoder contain more long-range information, which can improve the integrity of the large-scale objects such as "Asphalt" and reduce the intraclass variance of objects distributed in different regions from the global perspective.



(a) (b)

Fig. 18. Demonstration on the impact of the local-enhanced transformer encoder module on the PU dataset. (a) T-SNE visualization of LESSFormer without the local transformer encoder. (b) T-SNE visualization of LESSFormer.

Moreover, the t-stochastic neighbor embedding (SNE) algorithm is exploited to visualize the high-dimensional output features of LESSFormer with or without the local transformer encoder in Fig. 18, where different colors represent different classes. The samples from different categories in the t-SNE visualization of the proposed LESSFormer method are uniformly scattered while that of LESSFormer without the local transformer encoder are relatively blended. For instance, the samples from category 3, category 5, and category 7 are mixed in the results of LESSFormer without the local transformer encoder, making it hard to distinguish these categories. With the benefit of the local transformer encoder, the proposed LESSFormer method can remarkably maximize intraclass similarity and minimize interclass similarity, resulting in higher classification accuracy.

5) Model Complexity and Efficiency Analysis: To evaluate the complexity and efficiency of the proposed LESSFormer method, we list the trainable parameters and the number of arithmetic operations [i.e., Floating Point Operations (FLOPs)] of different DL-based methods on three datasets in Table XI, and report the training time as well as the test time of different classification methods on three datasets in Table XII. According to Table XI, LESSFormer requires fewer or comparable parameters compared to other methods, while its FLOPs is significantly higher than other methods except FreeNet. A reasonable interpretation is that the batch size was set to 1 for all DL-based methods when calculating FLOPs, and the LESSFormer as well as FreeNet fed the whole HSI image rather than a small cropped patch into networks for every batch. However, LESSFormer still requires fewer FLOPs than the patch-based methods when taking all patches (i.e., 800 000, 207 400, and 664 845 patches for the XA, PU, and HU datasets) into consideration. In addition, it is noticed

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TABLE XI
TRAINABLE PARAMETERS AND FLOPS OF DIFFERENT DL-BASED METHODS ON THREE DATASETS

Dataset	Complexity	HPDM-SPRN	MSRN	FreeNet	ViT	HiT	SpectralFormer	SSFTT	LESSFormer
Xiong'an	Params (K)	220.37	**139.73**	2845.59	210.13	57231.41	541.38	148.10	388.88
	FLOPs (M)	1743.82	40.10	513599.40	**1.05**	851.65	104.70	7.24	61730.82
Pavia University	Params (K)	184.63	**138.44**	2712.76	209.80	49181.06	209.19	147.78	200.15
	FLOPs (M)	1460.75	40.10	110718.42	**1.05**	591.16	26.04	7.24	20555.12
Houston University	Params (K)	193.27	**139.98**	2748.95	210.19	50656.85	270.96	148.17	374.32
	FLOPs (M)	1531.72	40.10	365132.04	**1.05**	639.00	42.42	7.24	52742.22

TABLE XII
RUNINNG TIME OF DIFFERENT CLASSIFICATION METHODS ON THREE DATASETS

Dataset	Running Time	SVM	EMP-SVM	HPDM-SPRN	MSRN	FreeNet	ViT	HiT	SpectralFormer	SSFTT	LESSFormer
Xiong'an	Train. (s)	**3.7**	141.4	131.3	51.7	77.4	133.4	427.0	80.4	11.4	820.8
	Test. (s)	31.7	162.8	238.8	234.3	18.7	17.8	2141.0	129.3	78.1	**1.3**
Pavia University	Train. (s)	**1.2**	17.3	44.0	21.3	197.0	12.3	213.7	301.2	11.7	121.0
	Test. (s)	3.3	17.7	25.0	6.0	6.3	1.5	142.3	65.4	5.9	**0.3**
Houston University	Train. (s)	45.8	281.6	419.7	523.8	65.8	**28.7**	2448.3	517.2	52.0	547.8
	Test. (s)	50.2	260.0	88.0	6.6	17.2	30.2	45.5	2.1	1.5	**0.9**

from Table XII that LESSFormer takes slightly more time for training compared with the other DL-based methods. However, the proposed model has the fastest inference speed on three datasets and takes 1.3, 0.3, and 0.9 s for testing on the large-scale XA, PU, and HU datasets, respectively. This might be explained by that the most time-consuming differentiable superpixel segmentation submodule is executed on the GPU for acceleration and we reduce the size of the assignment matrix $\mathbf{R}$ in this submodule from $N \times M$ to $N \times 9$ by only considering the nine surrounding superpixels from each pixel. Consequently, the proposed LESSFormer method is suitable for practical applications with a negligible test time cost.

V. CONCLUSION

In this article, we have proposed a novel LESSFormer method tailored for HSI classification, which embeds the properties of HSI data into the architecture design of vision transformer. Significantly different from preexisting approaches, LESSFormer can generate low-level spectral–spatial tokens tailored for HSI instead of yielding tokens from a single spectral or spatial dimension of the original HSIs. Moreover, LESSFormer is competent in modeling locality even when inadequate training samples are available, while the other transformer-based classification methods encounter difficulty in learning the implicit locality structures. In summary, the proposed LESSFormer method is able to develop the strengths and avoid the shortcomings of vision transformer to tackle the challenges in HSI classification task, by means of generating tokens specially for HSI data and building the local and long-range dependencies between tokens simultaneously. Extensive experiments on three benchmark HSI datasets illustrate the effectiveness and robustness of LESSFormer.

In the future, we will further explore the self-supervised variants of vision transformer to reduce the demand for training data by extracting valuable semantic features from a large amount of unlabeled HSI data.

REFERENCES

[1] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Apr. 2019.

[2] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[3] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[4] S. Lorenz, P. Ghamisi, M. Kirsch, R. Jackisch, B. Rasti, and R. Gloaguen, "Feature extraction for hyperspectral mineral domain mapping: A test of conventional and innovative methods," *Remote Sens. Environ.*, vol. 252, Jan. 2021, Art. no. 112129.

[5] Y. Chen, T.-Z. Huang, W. He, X.-L. Zhao, H. Zhang, and J. Zeng, "Hyperspectral image denoising using factor group sparsity-regularized nonconvex low-rank approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[6] H. Xu, W. He, L. Zhang, and H. Zhang, "Unsupervised spectral–spatial semantic feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[7] W. Zhao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Aug. 2016.

[8] X. He and P. Niyogi, "Locality preserving projections," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 16, 2003, pp. 1–8.

[9] S. Mika, G. Ratsch, J. Weston, B. Scholkopf, and K.-R. Mullers, "Fisher discriminant analysis with kernels," in *Proc. 9th Neural Netw. Signal Process., IEEE Signal Process. Soc. Workshop*, Aug. 1999, pp. 41–48.

[10] Y. Duan, H. Huang, and T. Wang, "Semisupervised feature extraction of hyperspectral image using nonlinear geodesic sparse hypergraphs," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[11] F. Luo, Z. Zou, J. Liu, and Z. Lin, "Dimensionality reduction and classification of hyperspectral image via multistructure unified discriminative embedding," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

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[12] X. Zhang, S. Shang, X. Tang, J. Feng, and L. Jiao, "Spectral partitioning residual network with spatial attention mechanism for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[13] Y. Liu, J. Li, and A. Plaza, "Spectrometer-driven spectral partitioning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 668–680, Feb. 2016.

[14] D. Hong, X. Wu, P. Ghamisi, J. Chanussot, N. Yokoya, and X. X. Zhu, "Invariant attribute profiles: A spatial-frequency joint feature extractor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3791–3808, Jun. 2020.

[15] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "A new deep convolutional neural network for fast hyperspectral image classification," *ISPRS J. Photogram. Remote Sens.*, vol. 145, pp. 120–147, Nov. 2018.

[16] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2017.

[17] S. Jia *et al.*, "3-D Gabor convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[18] B. Xi, J. Li, Y. Li, R. Song, W. Sun, and Q. Du, "Multiscale context-aware ensemble deep KELM for efficient hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5114–5130, Jun. 2021.

[19] P. Sellars, A. I. Aviles-Rivero, and C.-B. Schonlieb, "Superpixel contracted graph-based learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4180–4193, Jun. 2020.

[20] J. Jiang, J. Ma, and X. Liu, "Multilayer spectral-spatial graphs for label noisy robust hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2 pp. 839–852, Feb. 2022.

[21] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[22] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jan. 2015.

[23] R. Hang, Q. Liu, D. Hong, and P. Ghamisi, "Cascaded recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5384–5394, Aug. 2019.

[24] S. Hao, W. Wang, and M. Salzmann, "Geometry-aware deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2448–2460, Mar. 2021.

[25] Z. Zhong, J. Li, D. A. Clausi, and A. Wong, "Generative adversarial networks and conditional random fields for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 50, no. 7, pp. 3318–3329, Jul. 2020.

[26] J. Wang, F. Gao, J. Dong, and Q. Du, "Adaptive DropBlock-enhanced generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5040–5053, Jun. 2021.

[27] Y. Xu, B. Du, and L. Zhang, "Beyond the patchwise classification: Spectral-spatial fully convolutional networks for hyperspectral image classification," *IEEE Trans. Big Data*, vol. 6, no. 3, pp. 492–506, Sep. 2020.

[28] Z. Zheng, Y. Zhong, A. Ma, and L. Zhang, "FPGA: Fast patch-free global learning framework for fully end-to-end hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5612–5626, Aug. 2020.

[29] H. Zhang, J. Zou, and L. Zhang, "EMS-GCN: An end-to-end mixhop superpixel-based graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[30] Y. Ding, Y. Chong, S. Pan, Y. Wang, and C. Nie, "Spatial–spectral unified adaptive probability graph convolutional networks for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Sep. 23, 2021, doi: 10.1109/TNNLS.2021.3112268.

[31] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017.

[32] A. B. Hamida, A. Benoit, P. Lambert, and C. B. Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[33] S. Ghaderizadeh, D. Abbasi-Moghadam, A. Sharifi, N. Zhao, and A. Tariq, "Hyperspectral image classification using a hybrid 3D–2D convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7570–7588, 2021.

[34] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.

[35] S. Pande and B. Banerjee, "HyperLoopNet: Hyperspectral image classification using multiscale self-looping convolutional networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 422–438, Jan. 2022.

[36] C. Yu, R. Han, M. Song, C. Liu, and C.-I. Chang, "Feedback attention-based dense CNN for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[37] L. Mou and X. X. Zhu, "Learning to pay attention on spectral domain: A spectral attention module-based convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 110–122, Sep. 2020.

[38] R. Hang, Z. Li, Q. Liu, P. Ghamisi, and S. S. Bhattacharyya, "Hyperspectral image classification with attention-aided CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2281–2293, Mar. 2021.

[39] J. Feng *et al.*, "Attention multibranch convolutional neural network for hyperspectral image classification based on adaptive region search," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5054–5070, Jun. 2021.

[40] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[41] Z. Liu *et al.*, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[42] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2020, pp. 213–229.

[43] Z. Wang, X. Cun, J. Bao, W. Zhou, J. Liu, and H. Li, "Uformer: A general U-shaped transformer for image restoration," 2021, *arXiv:2106.03106*.

[44] M. Raghu, T. Unterthiner, S. Kornblith, C. Zhang, and A. Dosovitskiy, "Do vision transformers see like convolutional neural networks?" in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 34, 2021, pp. 1–13.

[45] Z. Zhao, D. Hu, H. Wang, and X. Yu, "Convolutional transformer network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[46] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[47] D. Hong *et al.*, "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[48] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Jan. 2020.

[49] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[50] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 30, 2017, pp. 1–11.

[51] B. Wu *et al.*, "Visual transformers: Token-based image representation and processing for computer vision," 2020, *arXiv:2006.03677*.

[52] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," 2016, *arXiv:1607.06450*.

[53] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," 2016, *arXiv:1606.08415*.

[54] V. Jampani, D. Sun, M.-Y. Liu, M.-H. Yang, and J. Kautz, "Superpixel sampling networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 352–368.

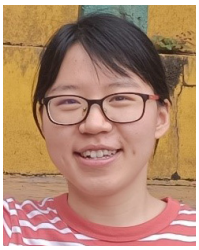
[55] Z. Fan *et al.*, "Mask attention networks: Rethinking and strengthen transformer," 2021, *arXiv:2103.13597*.

[56] A. Tan, D. T. Nguyen, M. Dax, M. Nießner, and T. Brox, "Explicitly modeled attention maps for image classification," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 11, pp. 9799–9807.

[57] H. Gao, Y. Yang, C. Li, L. Gao, and B. Zhang, "Multiscale residual network with mixed depthwise convolution for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3396–3408, Apr. 2021.

[58] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

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